

# Calculus and Analytical Geometry

## Equations of Lines and Planes

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## Objectives

After preparing this topic, you should be able to:

1. **Describe lines in three-dimensional space using vector, parametric, and symmetric equations, and use these representations to analyze lines and their relationships.**
2. **Describe planes in three-dimensional space using normal vectors and scalar equations, and use these equations to determine intersections, parallelism, and angles between planes.**

## The Why Section

### Objective 1: Why Study Equations of Lines?

- In three-dimensional engineering and computing applications, we often need to describe the position and direction of an object mathematically.
- For example, consider a robotic arm moving its end-effector through three-dimensional space. The robot may move along a straight path from one position to another, so we need a mathematical equation that describes every point on that path.
- A point alone tells us **where** the motion starts, but it does not tell us **which direction** the object moves.
- A direction vector together with a point completely determines a line in

space.

- Once the line is represented mathematically, we can determine the position of the object for different values of a parameter, find where its path intersects a plane, and determine whether two paths intersect, are parallel, or are skew.
- Therefore, vector, parametric, and symmetric equations provide mathematical tools for describing and analyzing straight-line motion in three-dimensional space.
- A robotic-arm positioning problem of this type can be solved by studying the first objective.

## Objective 2: Why Study Equations of Planes?

- In engineering and computer science, many important objects and boundaries are naturally represented by planes.
- For example, a computer-controlled cutting machine may move a tool toward a flat surface, while a robot may need to determine where a straight trajectory intersects a planar surface.
- To describe such a surface mathematically, we need more information than a single point.
- A vector perpendicular to the plane, called a **normal vector**, provides the direction that uniquely determines the orientation of the plane.
- Once a point on the plane and a normal vector are known, we can construct an equation that describes every point on the plane.
- Plane equations also allow us to determine whether two planes are parallel, find the angle between them, and determine the line where two nonparallel planes intersect.
- Therefore, equations of planes provide a mathematical framework for describing flat surfaces and analyzing their geometric relationships in three-dimensional space.
- A robotic-tool and planar-surface positioning problem of this type can be solved by studying the second objective.

# Objective 1: Lines

## Vector Equation of a Line

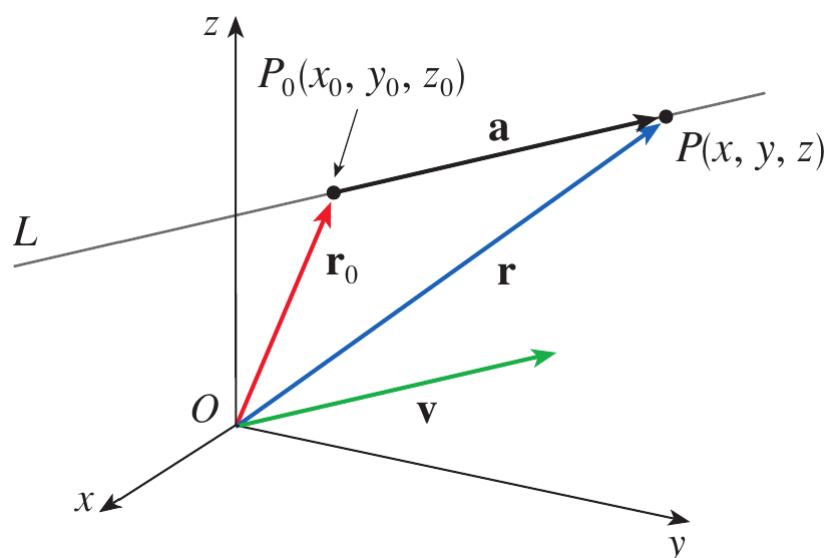
- A line in the  $xy$ -plane is determined when a point on the line and its direction are known.
- The same basic idea applies to a line in three-dimensional space.
- Suppose a line  $L$  passes through the point  $P_0(x_0, y_0, z_0)$  and has direction vector  $\mathbf{v}$ .
- Let  $\mathbf{r}_0$  be the position vector of  $P_0$  and let  $\mathbf{r}$  be the position vector of an arbitrary point  $P(x, y, z)$  on the line.
- The vector from  $P_0$  to  $P$  is parallel to the direction vector  $\mathbf{v}$ .
- Therefore, there is a scalar  $t$  such that

$$\mathbf{r} - \mathbf{r}_0 = t\mathbf{v}$$

- Hence, the vector equation of the line is

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$$

- As the parameter  $t$  varies through all real numbers, the tip of  $\mathbf{r}$  traces out the line.



**Figure 1.** A line in three-dimensional space determined by a point and a

direction vector.

- If the direction vector is written in component form as  $\mathbf{v} = \langle a, b, c \rangle$  and  $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$  then the vector equation becomes

$$\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t \langle a, b, c \rangle$$

- Equating corresponding components gives the parametric equations.

$$\langle x, y, z \rangle = \langle x_0 + at, y_0 + bt, z_0 + ct \rangle$$

## Parametric Equations of a Line

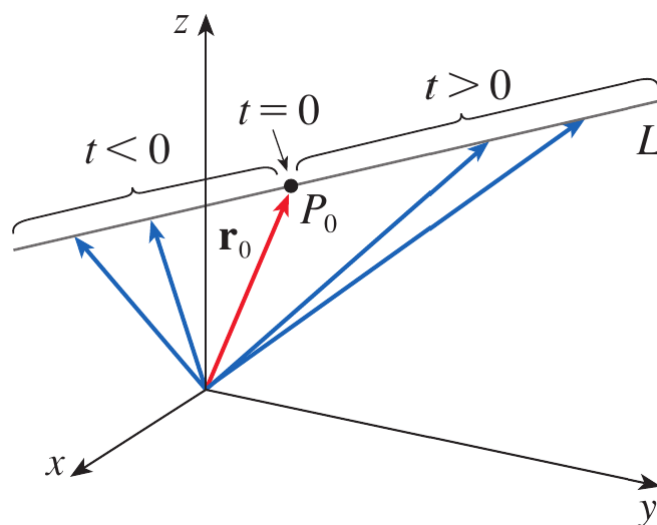
- The parametric equations of a line through  $P_0(x_0, y_0, z_0)$  parallel to the direction vector  $\langle a, b, c \rangle$  are

$$x = x_0 + at$$

$$y = y_0 + bt$$

$$z = z_0 + ct$$

- Every value of  $t$  gives a point on the line.
- Different choices of the starting point or a parallel direction vector can produce different-looking equations for the same line.



**Figure 2.** A line traced by the position vector as the parameter  $t$  varies.

## Example 1

Find a vector equation and parametric equations for the line that passes through the point  $(5, 1, 3)$  and is parallel to the vector  $\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ .

Find two other points on the line.

## Solution

The given point has position vector

$$\mathbf{r}_0 = 5\mathbf{i} + \mathbf{j} + 3\mathbf{k}$$

The direction vector is

$$\mathbf{v} = \mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$$

Therefore, using the vector equation of a line,

$$\mathbf{r} = (5\mathbf{i} + \mathbf{j} + 3\mathbf{k}) + t(\mathbf{i} + 4\mathbf{j} - 2\mathbf{k})$$

Equivalently,

$$\mathbf{r} = (5 + t)\mathbf{i} + (1 + 4t)\mathbf{j} + (3 - 2t)\mathbf{k}$$

Therefore, the parametric equations are

$$x = 5 + t$$

$$y = 1 + 4t$$

$$z = 3 - 2t$$

For  $t = 1$ ,

$$x = 6, \quad y = 5, \quad z = 1$$

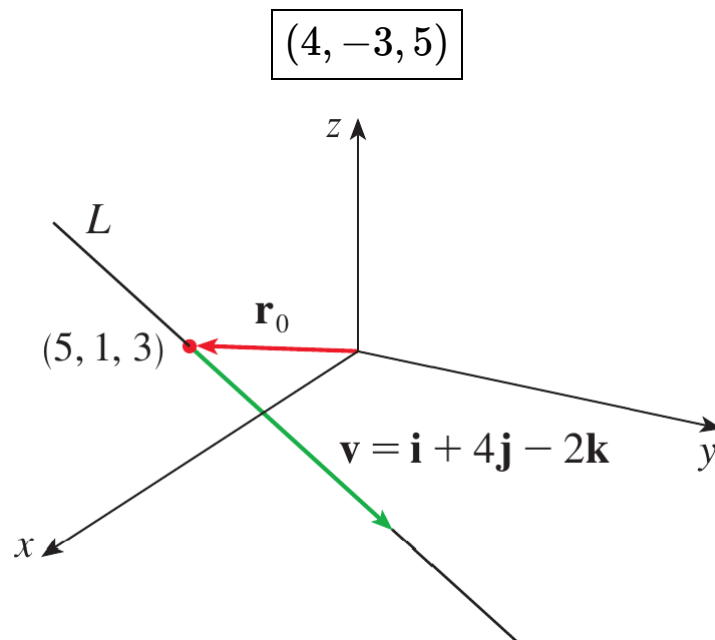
Thus, one other point on the line is

$$(6, 5, 1)$$

For  $t = -1$ ,

$$x = 4, \quad y = -3, \quad z = 5$$

Thus, another point is



**Figure 3.** A line and the given points used to describe its direction and position.

## Direction Numbers and Symmetric Equations

- If a direction vector of a line is  $\mathbf{v} = \langle a, b, c \rangle$  then  $a$ ,  $b$ , and  $c$  are called the **direction numbers** of the line.
- Consider the parametric equations

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

- From these equations, we can solve for  $t$  when the corresponding direction number is nonzero.

$$t = \frac{x - x_0}{a}$$

$$t = \frac{y - y_0}{b}$$

$$t = \frac{z - z_0}{c}$$

- Equating these expressions gives the symmetric equations of the line.

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

- If one of the direction numbers is zero, the corresponding coordinate remains constant rather than appearing as a denominator.

## Example 2

Find parametric equations and symmetric equations of the line that passes through the points  $A(2, 4, -3)$  and  $B(3, -1, 1)$ .

At what point does this line intersect the  $xy$ -plane?

## Solution

A direction vector for the line is obtained from  $\overrightarrow{AB}$ .

$$\mathbf{v} = \langle 3 - 2, -1 - 4, 1 - (-3) \rangle$$

$$\mathbf{v} = \langle 1, -5, 4 \rangle$$

Taking  $A(2, 4, -3)$  as the point on the line, the parametric equations are

$$x = 2 + t$$

$$y = 4 - 5t$$

$$z = -3 + 4t$$

Therefore, the symmetric equations are

$$\frac{x - 2}{1} = \frac{y - 4}{-5} = \frac{z + 3}{4}$$

To find where the line intersects the  $xy$ -plane, we use  $z = 0$ .

$$-3 + 4t = 0$$

Hence,

$$t = \frac{3}{4}$$

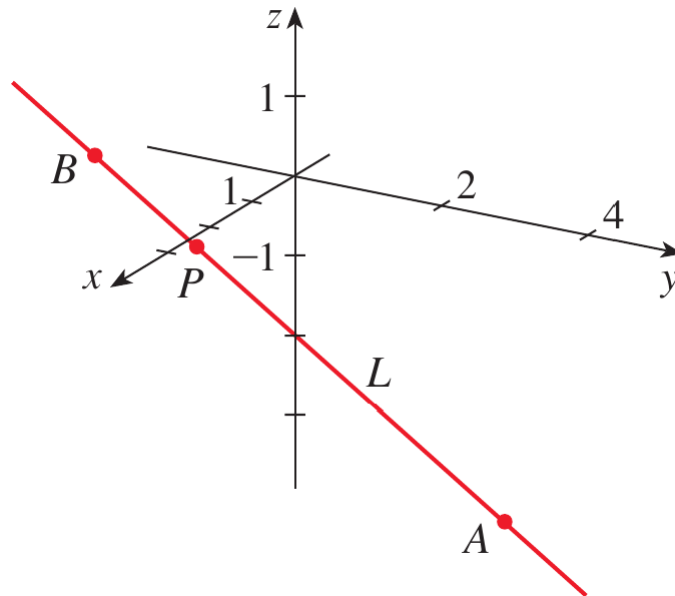
Substituting this value into the equations for  $x$  and  $y$  gives

$$x = 2 + \frac{3}{4} = \frac{11}{4}$$

$$y = 4 - 5 \left( \frac{3}{4} \right) = \frac{1}{4}$$

Therefore, the line intersects the  $xy$ -plane at

$$\left( \frac{11}{4}, \frac{1}{4}, 0 \right)$$



**Figure 4.** Line  $L$  and point  $P$  where it intersects the  $xy$ -plane.



## Line Segments

- A line extends indefinitely in both directions, but sometimes we need only the portion between two points.
- Suppose the position vectors of two endpoints are  $\mathbf{r}_0$  and  $\mathbf{r}_1$ .
- The line through the two points can be written as

$$\mathbf{r} = \mathbf{r}_0 + t(\mathbf{r}_1 - \mathbf{r}_0)$$

- For the line segment, the parameter is restricted to

$$0 \leq t \leq 1$$



- Therefore, the vector equation of the line segment from  $\mathbf{r}_0$  to  $\mathbf{r}_1$  is

$$\mathbf{r}(t) = (1 - t)\mathbf{r}_0 + t\mathbf{r}_1 \quad 0 \leq t \leq 1$$

## Relationships Between Lines

- Two lines in space can be **parallel**, **intersecting**, or **skew**.
- Two lines are parallel when their direction vectors are parallel.
- If their direction vectors are not parallel, we can determine whether they intersect by solving their parametric equations simultaneously.
- If two nonparallel lines do not intersect, they are called **skew lines**.

### Example 3

Show that the lines

$$L_1 : \quad x = 1 + t, \quad y = -2 + 3t, \quad z = 4 - t$$

$$L_2 : \quad x = 2s, \quad y = 3 + s, \quad z = -3 + 4s$$

are skew lines.

### Solution

The direction vectors are

$$\mathbf{v}_1 = \langle 1, 3, -1 \rangle$$

and

$$\mathbf{v}_2 = \langle 2, 1, 4 \rangle$$

These vectors are not scalar multiples of each other, so the lines are not parallel.

If the lines intersect, there must be values of  $t$  and  $s$  satisfying all three equations simultaneously.

Equating corresponding coordinates gives

$$1 + t = 2s$$

$$-2 + 3t = 3 + s$$

$$4 - t = -3 + 4s$$

Solving the first two equations gives

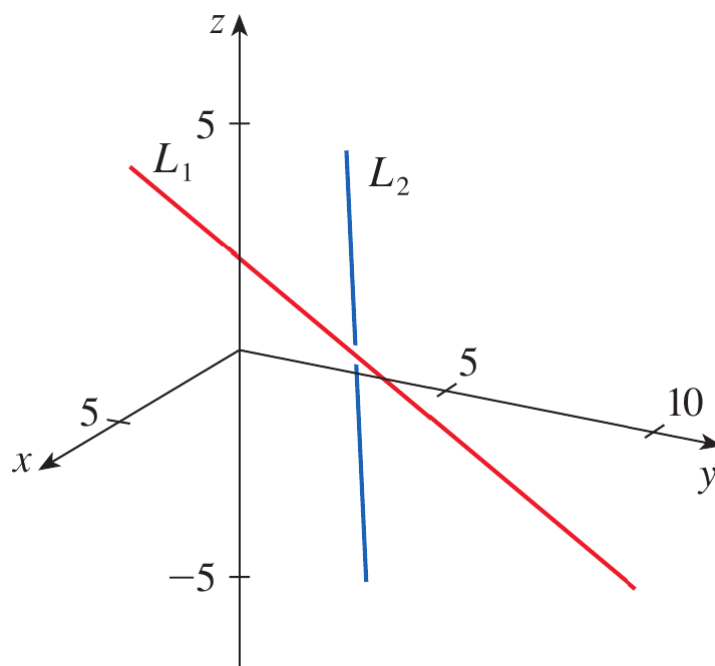
$$t = \frac{11}{5} \quad \text{and} \quad s = \frac{8}{5}$$

However, these values do not satisfy the third equation.

Therefore, the two lines do not intersect.

Since the lines are neither parallel nor intersecting, they are skew lines.

$L_1$  and  $L_2$  are skew lines



**Figure 5.** Two skew lines in three-dimensional space.



## Quick Check

### Question 1

Find parametric equations for the line through  $P(1, 2, -1)$  parallel to  $\mathbf{v} =$

$\langle 2, -3, 4 \rangle$ .

## Question 2

Are the lines

$$L_1 : \quad x = 1 + t, \quad y = 2 + 2t, \quad z = 3 - t$$

and

$$L_2 : \quad x = 4 + 2s, \quad y = 8 + 4s, \quad z = 1 - 2s$$

parallel?



## Objective 2: Planes

### Normal Vector to a Plane

- A line in space is determined by a point and a direction vector, but a plane requires a different type of directional information.
- A vector perpendicular to a plane completely determines the orientation of that plane.

### Definition: Normal Vector

A **normal vector** to a plane is a nonzero vector that is perpendicular to every direction lying in the plane.



- Suppose a plane contains the point  $P_0(x_0, y_0, z_0)$  and has normal vector

$$\mathbf{n} = \langle a, b, c \rangle$$

- Let  $P(x, y, z)$  be any point on the plane.
- The vector from  $P_0$  to  $P$  is

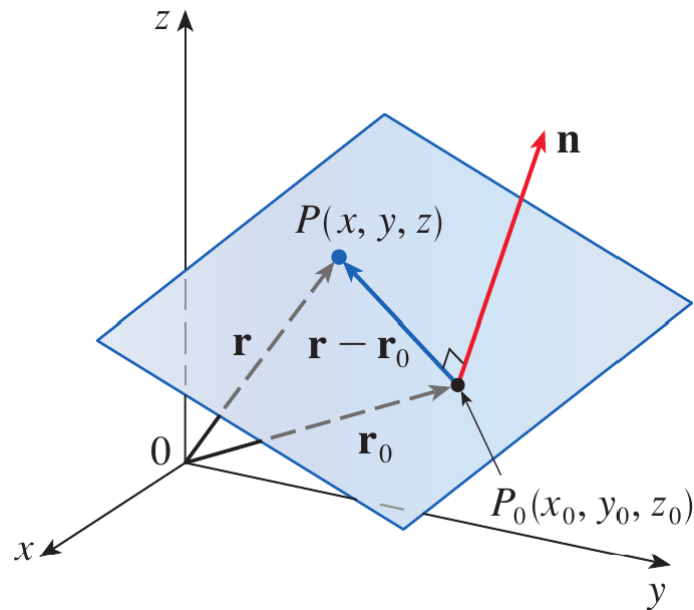
$$\mathbf{r} - \mathbf{r}_0$$

- Since this vector lies in the plane and  $\mathbf{n}$  is perpendicular to the plane, their dot product is zero.

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$$

- Equivalently,

$$\mathbf{n} \cdot \mathbf{r} = \mathbf{n} \cdot \mathbf{r}_0$$



**Figure 6.** A plane determined by a point and a normal vector.

## Scalar Equation of a Plane

- Let  $\mathbf{n} = \langle a, b, c \rangle$  and  $\mathbf{r} = \langle x, y, z \rangle$ .
- If  $P_0 = (x_0, y_0, z_0)$  then the vector equation  $\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$  becomes

$$\langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

- Taking the dot product gives the scalar equation of the plane.

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

- Expanding this equation gives the familiar linear form

$$ax + by + cz + d = 0$$

- Here  $\langle a, b, c \rangle$  is a normal vector to the plane.

## Example 4

Find an equation of the plane through the point  $(2, 4, -1)$  with normal vector  $\mathbf{n} = (2, 3, 4)$ . Find the intercepts and sketch the plane.

## Solution

The point is

$$(x_0, y_0, z_0) = (2, 4, -1)$$

The normal vector is

$$\mathbf{n} = \langle 2, 3, 4 \rangle$$

Using the scalar equation of a plane,

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

we obtain

$$2(x - 2) + 3(y - 4) + 4(z + 1) = 0$$

Expanding,

$$2x - 4 + 3y - 12 + 4z + 4 = 0$$

Therefore,

$$\boxed{2x + 3y + 4z = 12}$$

To find the  $x$ -intercept, set  $y = z = 0$ .

$$2x = 12$$

$$x = 6$$

Thus, the  $x$ -intercept is  $(6, 0, 0)$ .

Similarly, setting  $x = z = 0$  gives

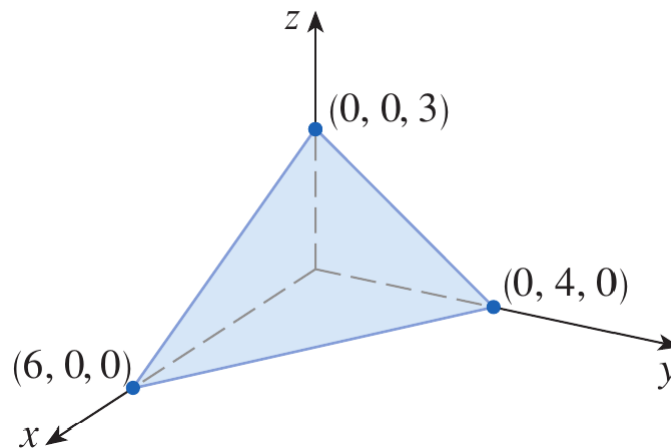
$$y = 4$$

Thus, the  $y$ -intercept is  $(0, 4, 0)$ .

Setting  $x = y = 0$  gives

$$z = 3$$

Thus, the  $z$ -intercept is  $(0, 0, 3)$ .



**Figure 7.** Plane through  $(2, 4, -1)$  with intercepts on the coordinate axes.



## Finding a Plane Through Three Points

- Three noncollinear points determine a unique plane.
- Suppose a plane contains the points  $P$ ,  $Q$ , and  $R$ .
- Two vectors lying in the plane can be formed as  $\mathbf{a} = \overrightarrow{PQ}$  and  $\mathbf{b} = \overrightarrow{PR}$ .
- Both vectors lie in the plane.
- Therefore, their cross product is perpendicular to both vectors and hence is a normal vector to the plane.

$$\mathbf{n} = \mathbf{a} \times \mathbf{b}$$

## Example 5

Find an equation of the plane that passes through the points

$$P(1, 3, 2), \quad Q(3, -1, 6), \quad R(5, 2, 0)$$

## Solution

First find two vectors lying in the plane.

$$\mathbf{a} = \overrightarrow{PQ}$$

$$\mathbf{a} = \langle 3 - 1, -1 - 3, 6 - 2 \rangle = \langle 2, -4, 4 \rangle$$

Similarly,

$$\mathbf{b} = \overrightarrow{PR}$$

$$\mathbf{b} = \langle 5 - 1, 2 - 3, 0 - 2 \rangle = \langle 4, -1, -2 \rangle$$

A normal vector is obtained from their cross product.

$$\mathbf{n} = \mathbf{a} \times \mathbf{b}$$

$$\mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -4 & 4 \\ 4 & -1 & -2 \end{vmatrix}$$

Expanding,

$$\mathbf{n} = 12\mathbf{i} + 20\mathbf{j} + 14\mathbf{k}$$

Therefore, a normal vector is

$$\mathbf{n} = \langle 12, 20, 14 \rangle$$

Using point  $P(1, 3, 2)$ , the equation of the plane is

$$12(x - 1) + 20(y - 3) + 14(z - 2) = 0$$

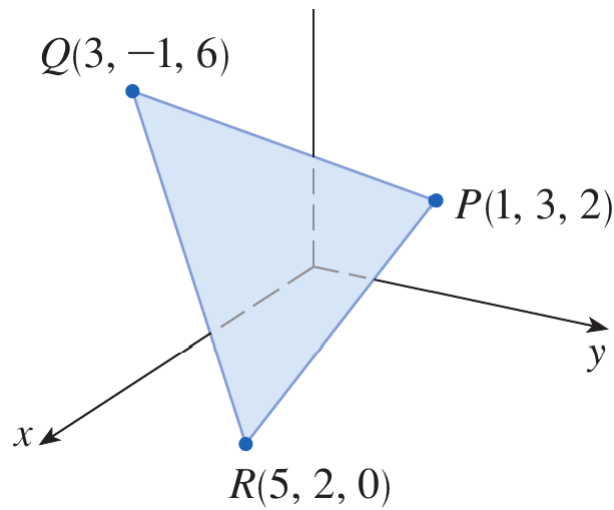
Dividing by 2 gives

$$6(x - 1) + 10(y - 3) + 7(z - 2) = 0$$

Therefore,

$$\boxed{6x + 10y + 7z = 50}$$

$z \uparrow$



**Figure 8.** Portion of the plane through the three points  $P$ ,  $Q$ , and  $R$ .

## Intersection of a Line and a Plane

- A line and a plane may intersect at one point, be parallel, or in special cases the line may lie entirely in the plane.
- When the line is given parametrically, we can substitute its coordinates into the equation of the plane.

### Example 6

Find the point at which the line with parametric equations

$$x = 2 + 3t, \quad y = -4t, \quad z = 5 + t$$

intersects the plane

$$4x + 5y - 2z = 18$$

### Solution

Substitute the parametric expressions for  $x$ ,  $y$ , and  $z$  into the plane equation.

$$4(2 + 3t) + 5(-4t) - 2(5 + t) = 18$$

Simplifying,



$$8 + 12t - 20t - 10 - 2t = 18$$

$$-10t - 2 = 18$$

$$-10t = 20$$

$$t = -2$$

Substitute  $t = -2$  into the parametric equations.

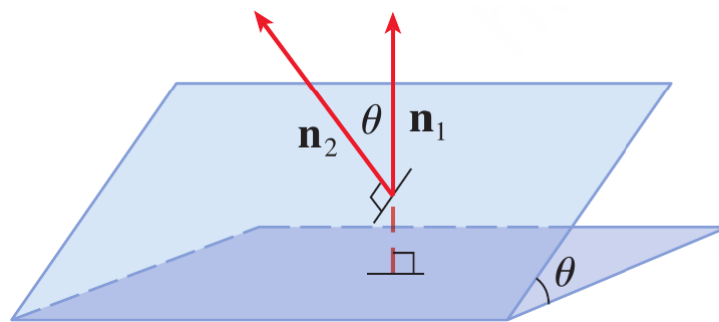
$$x = 2 + 3(-2) = -4$$

$$y = -4(-2) = 8$$

$$z = 5 + (-2) = 3$$

Therefore, the point of intersection is

$$\boxed{(-4, 8, 3)}$$



**Figure 9.** Intersection of line and plane.



## Relationships Between Planes

- The relationship between two planes can be determined from their normal vectors.
- If two normal vectors are parallel, the planes are parallel.
- For example, the planes  $x + 2y - 3z = 4$  and  $2x + 4y - 6z = 3$  have normal vectors  $\mathbf{n}_1 = \langle 1, 2, -3 \rangle$  and  $\mathbf{n}_2 = \langle 2, 4, -6 \rangle$ .
- Since  $\mathbf{n}_2 = 2\mathbf{n}_1$  the planes are parallel.

- If the normal vectors are not parallel, the planes intersect in a line.
- The angle between two planes is defined as the acute angle between their normal vectors.
- If  $\theta$  is the angle between the planes and  $\mathbf{n}_1$  and  $\mathbf{n}_2$  are normal vectors, then

$$\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|}$$

## Example 7

Find the angle between the planes

$$x + y + z = 1$$

and

$$x - 2y + 3z = 1$$

Find symmetric equations for the line of intersection  $L$  of these two planes.

## Solution

### Part (a): Angle Between the Planes

The normal vectors are

$$\mathbf{n}_1 = \langle 1, 1, 1 \rangle$$

and

$$\mathbf{n}_2 = \langle 1, -2, 3 \rangle$$

Their dot product is

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = 1(1) + 1(-2) + 1(3) = 2$$

Their magnitudes are

$$|\mathbf{n}_1| = \sqrt{3}$$

and

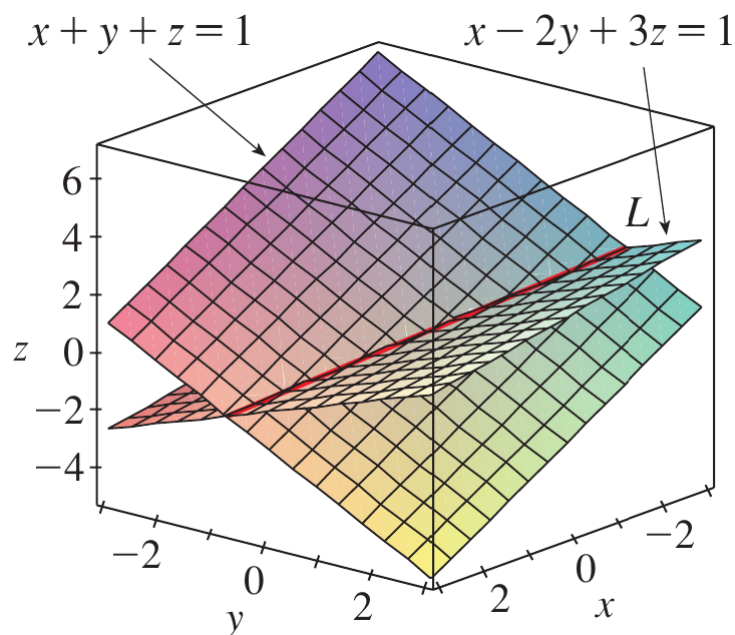
$$|\mathbf{n}_2| = \sqrt{1 + 4 + 9} = \sqrt{14}$$

Therefore,

$$\cos \theta = \frac{2}{\sqrt{3}\sqrt{14}} = \frac{2}{\sqrt{42}}$$

Hence,

$$\theta = \cos^{-1} \left( \frac{2}{\sqrt{42}} \right) \approx 72^\circ$$



**Figure 10.** Two planes with normal vectors showing the angle between the planes.

### Part (b): Line of Intersection

First find a point on the line of intersection.

Set  $z = 0$  in the two plane equations.

$$x + y = 1$$

$$x - 2y = 1$$

Solving these equations gives

$$x = 1, \quad y = 0$$

Thus, a point on the intersection line is

$$P = (1, 0, 0)$$

The line of intersection is perpendicular to both normal vectors.

Therefore, its direction vector is parallel to their cross product.

$$\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2$$

$$\mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix}$$

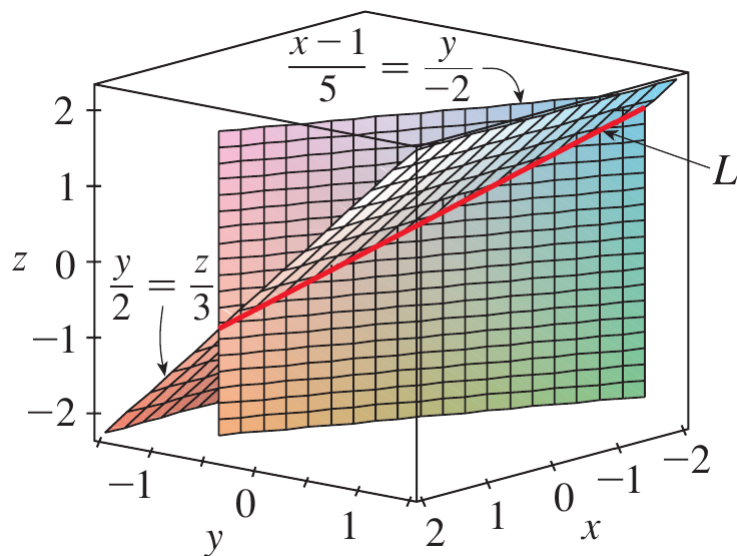
$$\mathbf{v} = 5\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$$

Thus,

$$\mathbf{v} = \langle 5, -2, -3 \rangle$$

Using the point  $(1, 0, 0)$  and this direction vector, the symmetric equations are

$$\frac{x-1}{5} = \frac{y}{-2} = \frac{z}{-3}$$



**Figure 11.** Two intersecting planes and their line of intersection.



# Quick Check

## Question 1

Find an equation of the plane through the point  $(1, 2, 3)$  with normal vector

$$\mathbf{n} = \langle 2, -1, 4 \rangle$$

## Question 2

Are the planes

$$2x + 4y - 6z = 5$$

and

$$x + 2y - 3z = 7$$

parallel?



# Scenario Problem

## Problem

A robotic positioning system moves the end-effector along a straight path in three-dimensional space. Its initial position is

$$P_0 = (1, 2, 3)$$

and its direction of motion is represented by

$$\mathbf{v} = \langle 2, -1, 2 \rangle$$

A flat calibration surface is represented by the plane

$$x + y + z = 12$$

Determine the point at which the robotic end-effector's straight-line path

intersects the calibration surface.

## Solution

The line passes through  $P_0 = (1, 2, 3)$  and has direction vector

$$\mathbf{v} = \langle 2, -1, 2 \rangle$$

Therefore, its parametric equations are

$$x = 1 + 2t$$

$$y = 2 - t$$

$$z = 3 + 2t$$

The calibration surface is

$$x + y + z = 12$$

Substitute the parametric expressions into the plane equation.

$$(1 + 2t) + (2 - t) + (3 + 2t) = 12$$

Simplifying,

$$6 + 3t = 12$$

Therefore,

$$t = 2$$

Substituting  $t = 2$  into the line equations gives

$$x = 1 + 2(2) = 5$$

$$y = 2 - 2 = 0$$

$$z = 3 + 2(2) = 7$$

Therefore, the robotic end-effector reaches the calibration surface at

$$\boxed{(5, 0, 7)}$$

This problem illustrates why equations of lines and planes are useful in

robotics and mechatronics: the line describes the robot's straight-line trajectory, while the plane describes the physical surface, and their intersection gives the exact position where the trajectory reaches the surface.



## Summary

- A line in three-dimensional space is determined by a point on the line and a direction vector.
- The vector equation of a line through  $\mathbf{r}_0$  with direction vector  $\mathbf{v}$  is  $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$ .
- Parametric equations describe the coordinates of points on a line as functions of a parameter  $t$ .
- Symmetric equations can be obtained from the parametric equations when the direction numbers are nonzero.
- Two lines in space may be parallel, intersecting, or skew, with skew lines being neither parallel nor intersecting.
- A plane is determined by a point on the plane and a normal vector perpendicular to the plane.
- The scalar equation of a plane through  $(x_0, y_0, z_0)$  with normal vector  $\langle a, b, c \rangle$  is  $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$ .
- Three noncollinear points determine a plane, and a normal vector can be obtained by taking the cross product of two vectors lying in the plane.
- Two planes are parallel when their normal vectors are parallel, while nonparallel planes intersect in a line whose direction is perpendicular to both normal vectors.
- Equations of lines and planes provide essential mathematical tools for describing trajectories, surfaces, intersections, and spatial relationships in engineering, robotics, computer graphics, and other three-dimensional applications.

# Exercises

## Exercises Set 1

Solve the odd number exercises from exercise 1 to 68.

- Exercises help transform theoretical concepts into practical understanding.
- Mathematics is learned by doing and solving exercises will train you to analyze problems, select appropriate methods, and construct logical solutions.
- Attempting problems sometimes leads to mistakes which provide opportunities for learning and improvement.
- Regular practice increases speed, accuracy, and confidence.
- Exercises are given in the Exercises file and you are expected to solve them on your own.